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Assignment - 01

Subject: Computer Networks

Subject Code: 24CST-339

Submitted By:

Name – Aditya Pratap Singh

UID – 24BAI70438

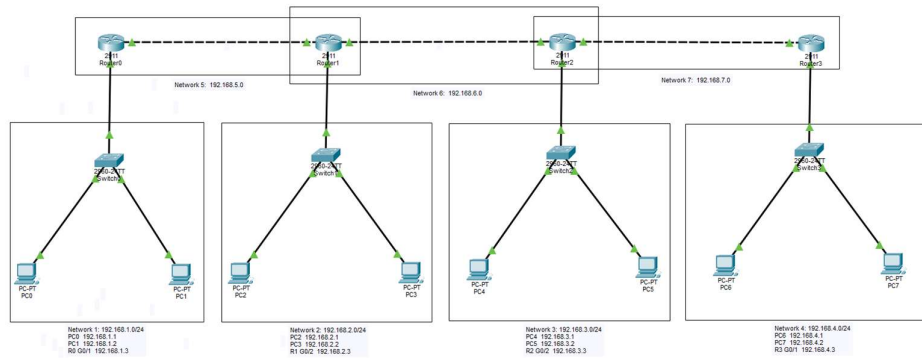
Section – 24AIT_KRG-1_B

Submitted To:

Faculty Name – Alok Sir

Q1. Develop Static Routing with 4 router's, 4 Switches & 8 PC's. Make sure to highlight the Networks with their network ID's. IP Addresses should be Written in text format as well.

Solution:



Network ID	Purpose	Devices Connected
192.168.1.0/24	LAN	Router0, Switch0, PC0, PC1
192.168.2.0/24	LAN	Router1, Switch1, PC2, PC3
192.168.3.0/24	LAN	Router2, Switch2, PC4, PC5
192.168.4.0/24	LAN	Router3, Switch3, PC6, PC7
192.168.5.0/24	WAN	Router0 ↔ Router1
192.168.6.0/24	WAN	Router1 ↔ Router2
192.168.7.0/24	WAN	Router2 ↔ Router3

Network	Device	Interface	IP Address
Network 1	PC0	FastEthernet0	192.168.1.1
	PC1	FastEthernet0	192.168.1.2
	Router0	G0/1	192.168.1.3
Network 2	PC2	FastEthernet0	192.168.2.1
	PC3	FastEthernet0	192.168.2.2
	Router1	G0/2	192.168.2.3
Network 3	PC4	FastEthernet0	192.168.3.1
	PC5	FastEthernet0	192.168.3.2
	Router2	G0/2	192.168.3.3
Network 4	PC6	FastEthernet0	192.168.4.1
	PC7	FastEthernet0	192.168.4.2
	Router3	G0/1	192.168.4.3
Network 5	Router0	Router link	192.168.5.1
	Router1	Router link	192.168.5.2
Network 6	Router1	Router link	192.168.6.1
	Router2	Router link	192.168.6.2
Network 7	Router2	Router link	192.168.7.1
	Router3	Router link	192.168.7.2

Router	Destination Network	Subnet Mask	Next Hop

Router0	192.168.2.0	255.255.255.0	192.168.5.2
	192.168.3.0	255.255.255.0	192.168.5.2
	192.168.4.0	255.255.255.0	192.168.5.2
Router1	192.168.1.0	255.255.255.0	192.168.5.1
	192.168.3.0	255.255.255.0	192.168.6.2
	192.168.4.0	255.255.255.0	192.168.6.2
Router2	192.168.1.0	255.255.255.0	192.168.6.1
	192.168.2.0	255.255.255.0	192.168.6.1
	192.168.4.0	255.255.255.0	192.168.7.2
Router3	192.168.1.0	255.255.255.0	192.168.7.1
	192.168.2.0	255.255.255.0	192.168.7.1
	192.168.3.0	255.255.255.0	192.168.7.1

Q2. Write the difference between Hub, Switch & Router in Tabular Format. Explain why router is the most intelligent & Hub is least intelligent device.

Solution:

Feature	Hub	Switch	Router
OSI Layer	Physical layer (Layer 1)	Data Link layer (Layer 2)	Network layer (Layer 3)
Working principle	Broadcasts incoming data to all ports	Forwards data only to the destination port using MAC addresses	Forwards data between different networks using IP addresses
Addressing used	None	MAC address	IP address

Collision domain	Single (shared for all ports)	Separate for each port	Separate for each port/network
Broadcast domain	Single	Single	Separate for each network
Intelligence	None — pure signal repeater	Learns MAC addresses, makes forwarding decisions	Reads IP headers, makes routing decisions, runs routing algorithms
Data unit handled	Bits	Frames	Packets
Typical use	Small, outdated LAN setups	Connecting devices within the same LAN	Connecting different networks (e.g. LAN to WAN/Internet)
Speed/efficiency	Slow, high collisions	Faster, fewer collisions	Slightly slower per-packet (due to processing) but efficient across networks

Why the router is the most intelligent device

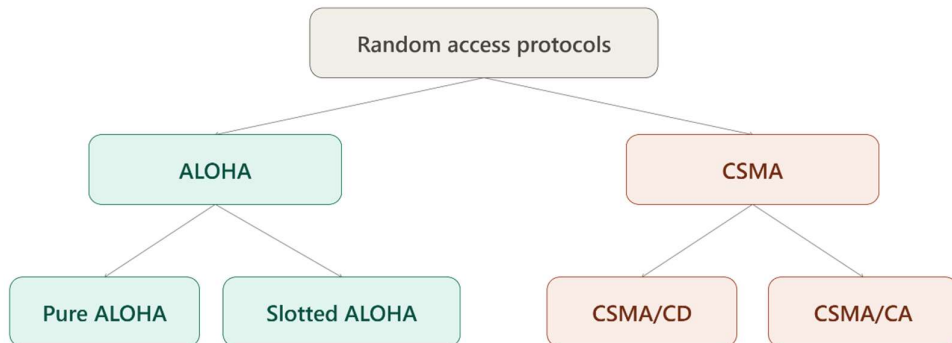
A router doesn't just move data — it thinks about where data should go. It reads the destination IP address in every packet, consults a routing table, and can choose among multiple possible paths using routing algorithms (like OSPF or RIP). It can connect entirely different networks, translate between them (via NAT), filter traffic, and make dynamic decisions based on network conditions. This decision-making capability is what makes it "intelligent."

Why the hub is the least intelligent device

A hub has zero awareness of what it's sending or to whom. When one device sends data, the hub simply copies that electrical signal out to every other connected port — it doesn't check addresses, doesn't know which port the destination device is on, and doesn't manage collisions. It's a pure signal amplifier/repeater, not a decision-maker. That's why all devices on a hub share a single collision domain — the hub can't isolate traffic at all.

Q3. What are Random Access Protocols? Why do we need them? Explain each category in detail with diagram and example. Also Explain difference between CSMA/CD & CSMA/CA.

Solution:



What they are: In random access protocols, there's no fixed schedule or turn for stations to transmit. Any station can send data whenever it has something to send — there's no central controller assigning time slots or permission. Because of this, two or more stations may transmit at the same time, causing a **collision**, and the protocol must define rules for handling that.

Why we need them: In a shared medium (like a common cable or wireless channel), if every device just transmitted whenever it wanted with no rules at all, collisions would be frequent and the network would become unusable. Random access protocols provide a structured way to reduce collisions, detect them when they happen, and recover — giving all stations fair, decentralized access to the channel without needing an expensive central controller (unlike controlled access methods such as polling or token passing).

Random access protocols fall into two main families, shown above: **ALOHA** and **CSMA**.

1. ALOHA

a) Pure ALOHA

A station transmits data the moment it has it ready — no checking whether the channel is free. If two stations transmit within a small time window of each other, their frames collide and are destroyed. The sender waits for an acknowledgment; if none arrives within a timeout, it assumes a collision occurred and retransmits after a random backoff time.

Example: Station A sends a frame at $t=2s$ and Station B sends at $t=2.3s$. Since their transmission times overlap, both frames collide and must be resent.

Vulnerable time = $2 \times T_t$ (frame transmission time), so throughput is low (maximum $\sim 18.4\%$).

b) Slotted ALOHA

An improvement on Pure ALOHA: time is divided into discrete slots equal to the frame transmission time, and a station is allowed to start transmission only at the beginning of a slot. This cuts the vulnerable period in half compared to Pure ALOHA, since a collision can now only happen if two stations pick the exact same slot.

Example: If slots are 1 second wide, a station can only begin sending at $t=0, 1, 2, 3...$ not at $t=0.5$.

Vulnerable time = T_t , giving better throughput (maximum $\sim 36.8\%$).

2. CSMA (Carrier Sense Multiple Access)

Here, a station **listens to the channel before transmitting** ("sense before send"). If the channel is busy, it waits; if idle, it transmits. This reduces collisions compared to ALOHA but doesn't eliminate them completely — because of propagation delay, two stations far apart might both sense the channel as idle and transmit at nearly the same time.

CSMA has two important refinements:

CSMA/CD (Collision Detection)

Used mainly in **wired networks** (traditional Ethernet). The station listens while transmitting. If it detects a collision (by sensing abnormal voltage/signal on the line), it immediately stops transmitting, sends a short jam signal to inform other stations, and retransmits after a random backoff period.

Example: Two computers on an old Ethernet LAN both start sending at once; each detects the collision within microseconds, stops immediately, and retries later — saving bandwidth compared to sending the whole frame.

CSMA/CA (Collision Avoidance)

Used mainly in **wireless networks** (Wi-Fi/802.11), because collision *detection* isn't practical over radio (a wireless device can't easily listen and transmit at the same time, and signal strength issues make it hard to detect collisions reliably). Instead, it tries to *avoid* collisions before they happen using techniques like:

- **Interframe spacing (IFS):** waiting a specific gap before sensing the channel.
- **Contention window / random backoff:** waiting a random time even after the channel is sensed idle.
- **RTS/CTS handshake:** Request to Send / Clear to Send control frames reserve the channel before actual data is sent.
- **ACK frames:** confirm successful receipt of each frame.

Difference between CSMA/CD and CSMA/CA

Aspect	CSMA/CD	CSMA/CA
Full form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Used in	Wired LANs (Ethernet)	Wireless LANs (Wi-Fi)
Approach	Detects collision after it occurs, then stops and retransmits	Tries to prevent collision before it occurs
Mechanism	Monitors the channel while transmitting	Uses IFS, random backoff, and RTS/CTS before transmitting



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Collision handling	Sends jam signal, aborts transmission	Reserves channel in advance to avoid collision altogether
Efficiency in wireless	Not usable — can't reliably detect collisions on radio	Well-suited to wireless medium